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Such a motion is possible.

(Option A is incorrect)

In a binary star system, the centre of mass of two stars of unequal mass will not be at the midway point. It will be closer to star with mass M . Hence the radius of orbit of M is smaller than $2M$.

(Option C is incorrect)

For an orbit centred around its common centre of mass, position of centre of mass is fixed in space. For this condition to be satisfied, the two stars have to constantly be on opposite sides of the common centre of mass. Hence, they will have the same period of rotation and hence same angular velocity.

(Option D is the answer)

Since they have the same angular velocity and different radius, the speed at which each star orbits will also be different from $v = r\omega$.

(Option B is incorrect)

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